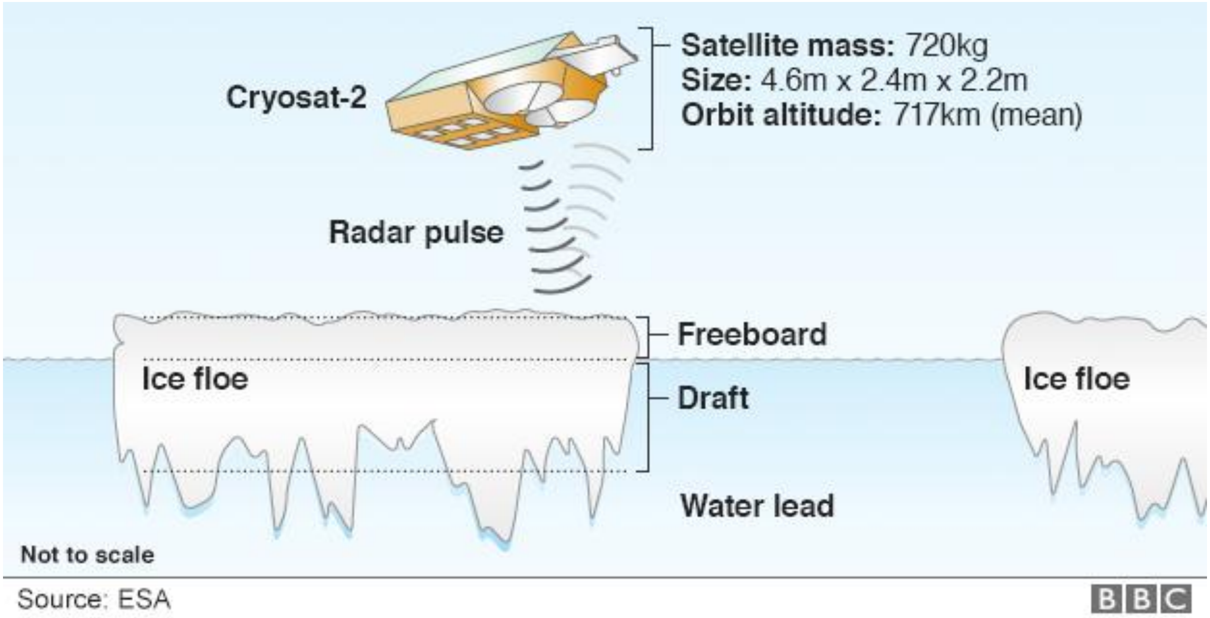
Tutorial-06
Cryosat-2 problem specification - iteration

Given:



The radar senses the height of the freeboard - the ice above the waterline – it is the top 1/9th of the floe. The snow and ice on the top of the freeboard is rarely a flat top surface and usually has troughs and peaks.

Knowing this 1/9th figure allows Cryosat to work out sea-ice thickness

Some 8/9ths of the ice tends to sit below the waterline - the draft

The thickness multiplied by the area of ice cover produces a volume

TASK: Pseudocode the following algorithm:
Accept three different readings for the freeboard and calculate the average.
(Use iteration to capture the readings)

If the average is below a certain threshold then discard the reading as being statistically insignificant else continue with the calculation of volume.

Output the freeboard, the draft, the area, the thickness, the volume

It has been decided to apply the above algorithm to any number of floes.